UCDAVIS VETERINARY MEDICINE

Urinary System Disorders

Hugues Beaufrère

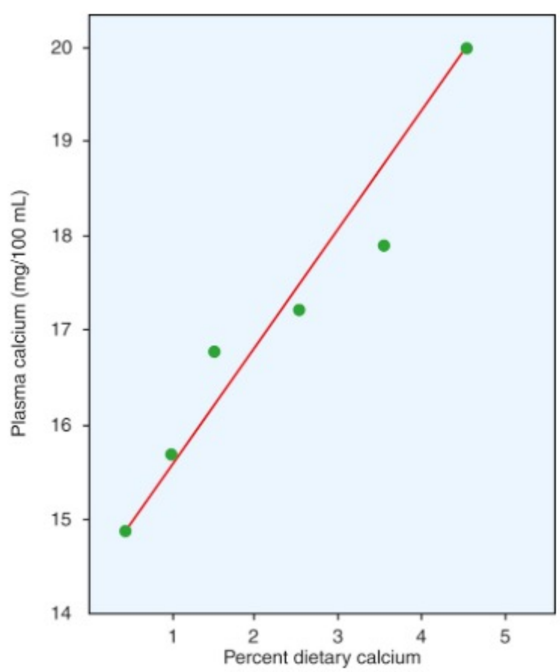
Diseases of the urinary system are common in small mammals, in particular in rabbits, guinea pigs, and ferrets. The diagnosis and management of these diseases is very close to what you have learned in domestic carnivores (dogs and cats). So you can extrapolate a lot keeping in mind a few important species-specific features. There are some unique aspects to the pathophysiology, diagnosis, and treatment of urinary tract disorders of small mammals that you should know and that will be highlighted here. For this reason, we won't necessarily be thorough in the description of the diagnostic workup of urinary diseases and the description of some diseases and we will mainly focus on differences and big take-home messages to help you apply and adapt your general knowledge to these species.

Likewise, we won't cover every single urinary tract disease that has been reported in rabbits, rodents, and ferrets, because some are quite rare and only reported a handful of times. It is a better use of your time to focus on what is common or unique about this system in small exotic mammals, so you can quickly generate a working differential diagnosis that will work in the majority of clinical scenarios. We will especially focus on urolithiasis as it has many unique aspects compared to domestic carnivores and primary renal diseases as they are also commonly seen.

RECAP: RABBIT AND RODENT CALCIUM METABOLISM

To fully understand this section and how it differs from domestic carnivores, it is necessary to go back a bit and discuss rabbit (and some rodents) calcium metabolism. You'll see that it will impact the diagnosis, pathophysiology, and treatment of a number of diseases of rabbits (and some rodents). Also, this makes such good materials for exams!!

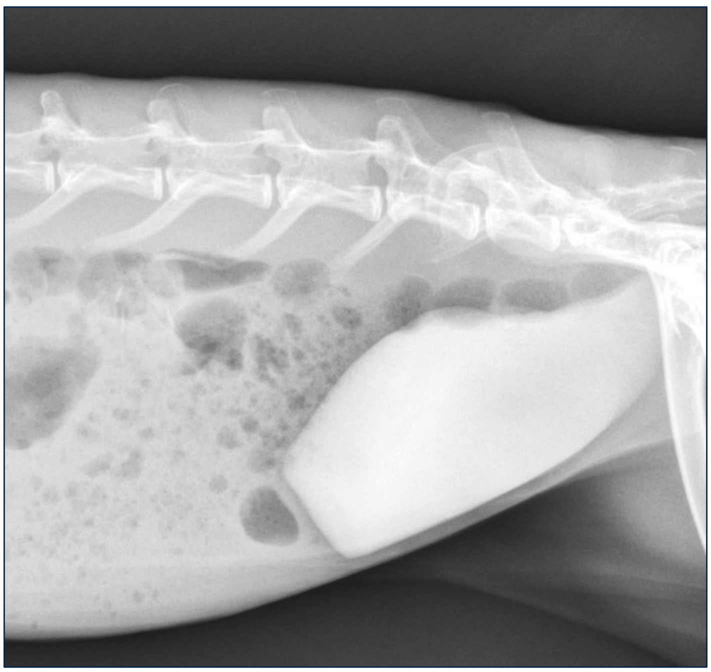
What makes rabbit unique (apart from their long ears and propensity to eat poop) is their calcium metabolism. **Rabbits have an amazing ability to absorb dietary calcium in the intestines. In fact, the absorption is mainly passive and unregulated by vitamin D** (whereas it regulates intestinal absorption in most other mammals). **As a result, plasma calcium is linearly related to dietary calcium.**



Guinea pigs also absorb more calcium than most other mammals, but intestinal absorption is still somewhat regulated by vitamin D.

The other thing that rabbits and guinea pigs do is that they massively excrete calcium in their urine. The urinary fractional excretion (amount that leaves the body compared to amount reabsorbed by kidneys) of calcium is a whopping 45-50% in them (compared to like a meagre 2% in most other mammals where calcium is mainly excreted in feces). In other rodents, such as rats and chinchillas, the urinary fractional excretion of calcium is also 2%.

So what are the clinical applications of this great knowledge that you'll remember all your life. First, **rabbits will have much higher total and ionized calcium than most other mammals. It is in fact something like 30-50% higher. Ionized calcium is the highest of any mammal** I know at 1.7-1.8 mmol/L (vs 1.2 in most mammals). There is a much higher threshold in rabbits to trigger negative PTH feedback. You should not misinterpret plasma calcium levels in rabbits for pathologic hypercalcemia. Likewise, you should recognize that what would be normal in another mammal (iCa of 1.2 mmol/L), is actually low in rabbits and could lead to treatment. Secondly, because of the high urinary excretion of calcium, it will lead to **very thick urine, often called "sludge". This is the result of a large amount of precipitates of calcium carbonate in alkaline urine in both rabbits and guinea pigs**.



This may look impressive, but it is a physiological process that indicates a high dietary calcium intake.

THE DIAGNOSTIC WORKUP

Reasons of Presentation

Most of the time, the owners bring you their pets with diseases of the urinary system for **non-specific signs such as chronic weight loss, lethargy, and decreased appetite**. More specific signs can be related to **urination disorders (incontinence, inappropriate urination, pollakiuria, stranguria), change in urine color (e.g. hematuria, porphyrins), decreased urinary function (PUPD, anuria), or abdominal pain (with a ureteral calculus for instance).** Owners may also present these animals for secondary issues not suspecting they are related to a urinary tract disease. These include **perineal soiling (= urine scalding, or "hutch burn") or gastrointestinal stasis (from dehydration, anorexia, abdominal pain).**

Physical Examination

On your complete physical examination, you may witness the following with urinary tract disorders:

- **Decreased body condition**: mainly in chronic renal disease or other chronic conditions such as chronic cystitis and urolithiasis.
- **Dehydration**: from decreased renal function, but also decreased fluid intake coupled to electrolytic loss.
- **Perineal dermatitis** as mentioned above.

- Abdominal palpation abnormalities such as **abnormal kidneys** (in shape or size with things like chronic renal disease, renal masses, renal cysts, hydronephrosis and so on), a **turgid bladder** (with urolithiasis or upper motor neuron bladder), or a **large cystic calculus** (those can often be palpated if large enough).

You should also be on the lookout for signs commonly associated with systemic disorders that also affect the urinary tract. For instance, encephalitozoonosis presents with a variety of **neurological signs** (mainly head tilts), but renal lesions are also common with that disease. Systemic neoplasias such as **lymphoma** may lead to organomegaly and masses and it is not uncommon to find lymphoma in the kidneys too. In ferrets, hair loss and other signs are associated with **adrenal gland disease**, which can result in urethral obstruction in males from prostatic disorders.

Likewise, **primary urinary tract disorders may affect other organ system** such as the skin (urine scald), the gastrointestinal system (gastric stasis), the cardiac system (hyperkalemia, hypo and hypertension), the hematopoietic system (anemia), the musculoskeletal system (weight loss from protein loss and muscle calcification), the neurological system (severe azotemia), and the endocrine system (PTH, Vit D, EPO, renin).

Bottom line, do not have tunnel vision and have a holistic approach to your patients.

Blood Work

The interpretation of the renal panel is **almost identical to the other mammals.** Urea and creatinine are both markers of glomerular filttration and can be elevated in pre-renal, renal, and post-renal azotemia. Creatinine is less influenced by external factors (such as diet, renal reabsorption) than urea, so creatinine is seen as a more reliable marker. When renal azotemia develops, there is already a substantial decrease in nephrons and renal function. SDMA has also been used, but there is less data on that analyte in our species.

Likewise, electrolytic changes, apart from calcium in rabbits and guinea pigs, are interpreted as in other mammals. Hyperphosphatemia and hyperkalemia are some changes that are frequently seen with urinary tract disorders.

On the CBC, depending on the underlying disease process, you may see **anemia** (because EPO is produced by the kidneys) and/or **leucocytosis** (with inflammatory and infectious disorders). Anemia is also quite common in older rabbits and is also seen with urolithiasis if significant chronic bleeding or cystitis is present, in particular in guinea pigs. Leucocytosis is not a common finding in rabbits, overall. Milder changes indicative of inflammation are more commonly observed such as heterophilia, an increased heterophil-to-lymphocyte ratio, or the presence of a left shift.

Urinalysis

Collecting urine from small mammals is slightly more difficult because of their size. It is **typically done by ultrasound-guided cystocentesis using a 25g needle**. However, urine can also be collected on the floor of the cage (make sure there is no substrate), by manual expression (be gentle) or urinary catheterization (mainly done in male rabbits, may have to be more heavily sedated). If you plan on doing a urinary culture, then it has to come from a cystocentesis to avoid contamination.

Urine of rabbits and guinea pigs **can appear extremely cloudy because of the large amount of calcium crystals/"sludge"**, which is normal and a direct reflection of their diet.

Calcium carbonate crystals tend to form in alkaline urine. It may be a good idea to centrifuge prior to analysis if it is very cloudy.

In rabbits and rodents, the urine color can change from yellow to dark brown because of the excretion of porphyrins. It is thought that these porphyrins are coming from the break down of endogenous compounds or ingested plant pigments. Porphyrinuria is common, is not pathologic, and should not be confused with hematuria.

To differentiate porphyrinuria from hematuria, you can do a dipstick (which should be negative with porphyrins, but the color may interfer with interpretation), look for blood clots in the urine, look for increased number of RBCs on the urinalysis, and use a Wood's lamp to look for fluorescence (with some porphyrins). In fact, the color is more like dark orange-brownish than bright red like frank blood.

Below are the normal specific gravity of a few small mammals. Note that **rabbits' urine is frequently hypo- or isosthenuric as they have limited concentration ability**. So you need to interpret the USG in context (it should be higher in a dehydrated animal or you should suspect renal disease).

	USG
Rabbit	1.005-1.036
Guinea pig	1.005-1.050
Chinchilla	1.014->1.060
Rat	1.022-1.050
Ferret	1.026-1.070 (higher in males)

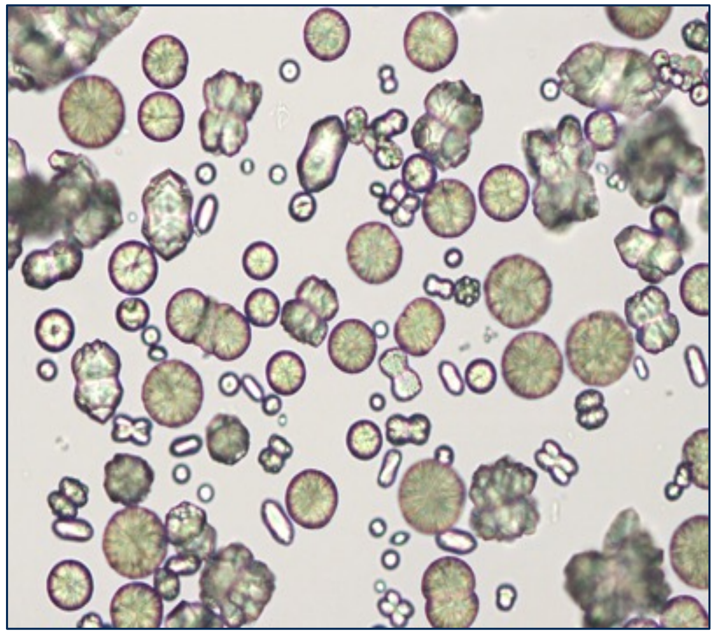
The pH is alkaline in herbivorous small mammals (rabbits and guinea pigs) whereas it is acidic in carnivorous small mammals (ferrets).

	pH
Rabbit	8-9
Guinea pig	8-9
Chinchilla	8.5-9.5
Rat	7.3-8.5
Ferret	5-6.5

For this reason, they don't tend to have the same calculi and urolithiasis management will obviously be different.

Other urinary dipstick parameters are typically negative. **Mild glycosuria may be seen in stressed individuals (particularly in rabbits). A 1+ ketones may be normal in some species of rodents. The protein and WBC pads are not reliable with high pH urine, so in rabbits and rodents.**

On evaluation of the sediment, **calcium carbonate crystals** are normal.



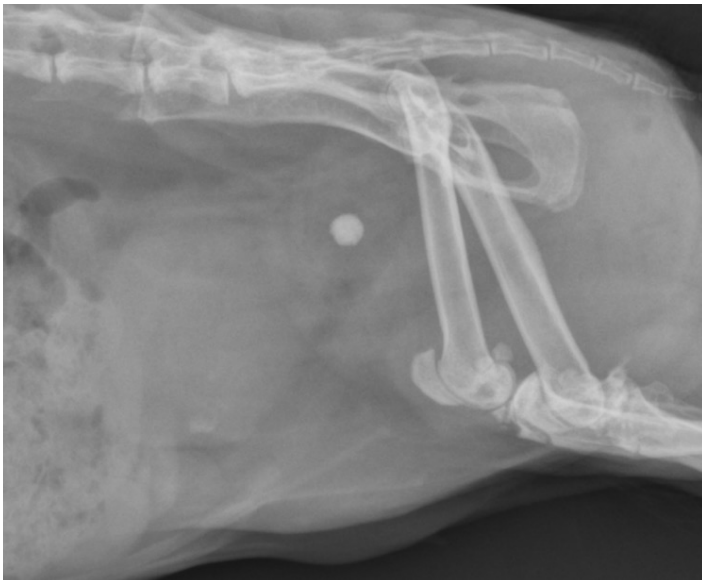
Other crystals may also be normally observed such as **ammonium magnesium phosphate**. No cast should be seen or it is indicative of renal disease. Only rare WBC, erythrocytes, and epithelial cells should be seen.

As in other mammals, a urinary protein to creatinine ratio (UPC) can be performed in small mammals.

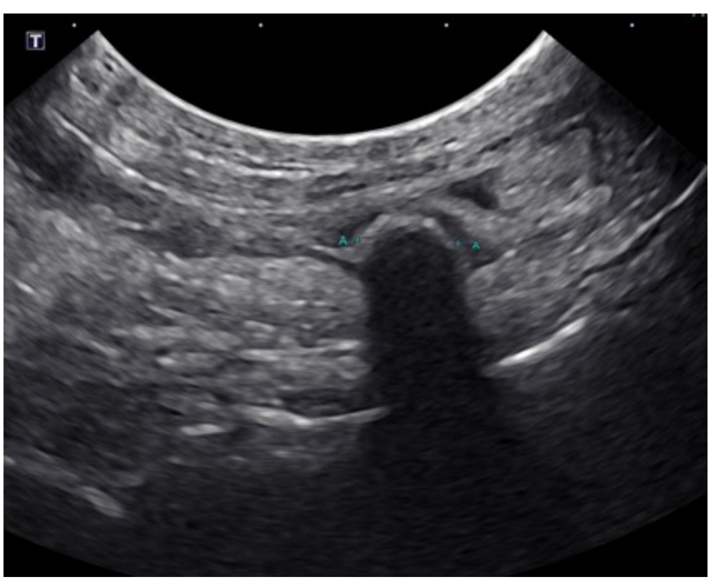
Diagnostic Imaging

Diagnostic imaging of the urinary tract mainly involves abdominal radiographs and abdominal ultrasound. The urinary tract can be seen very well on CT-scan, but it does not necessarily add much more than a regular ultrasonographic examination. The only thing that is nice with CT is to perform an **excretory urogram** with intravenous iodinated contrast agents and make sure the kidneys work and track the ureters. I am not going to dwell too much on diagnostic imaging as it is essentially identical to similar procedures in domestic carnivores.

On radiographs, you can mainly assess the kidneys and the bladder. It is particularly useful to screen for radio-opaque materials such as uroliths in the kidneys, ureters, bladder, or urethra. **You should always perform survey radiographs** as you can miss some calculi on ultrasound. **Note that some calculi like cystine calculi are mainly radiolucent (mainly an issue in ferrets).**



An ultrasound examination is really the best type of imaging modalities to evaluate the entire urinary tract. The thickness and vascularization of the bladder wall is a good indication of severe cystitis, commonly seen in guinea pigs. You can also evaluate the prostate in male ferrets (see later for why this is important). Ultrasound-guided cystocensis is the best method of urine collection and can be performed right after the examination. Lastly, you can do ultrasound-guided fine-needle aspiration of lesions such as renal nodules or other masses. The image below is of a cystic calculus in a guinea pig.



UROLITHIASIS

Rabbits and Rodents

Urolithiasis is very very common in rabbits and guinea pigs and present slightly differently from dogs and cats. It is a bit less common in chinchilla and is rare in rats.

The causes are not fully understood, but it seems to be associated with a combination of **high calcium diet (such as alfalfa hay or pellets, greens high in calcium such as kale and parsley) and high calcium urinary excretion in this species. Obesity and lack of exercise are risk factors.** Male rabbits are over-represented compared to females while there is no sex predisposition in guinea pigs. **Urolithiasis is not commonly associated with bacterial infection, but can, especially in guinea pigs.**

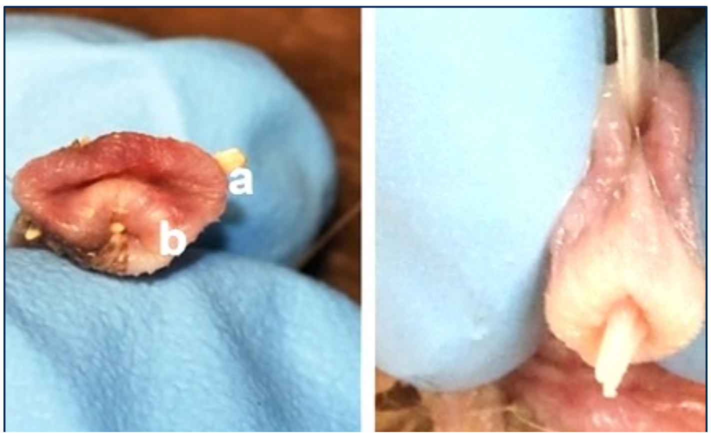
The vast majority of uroliths in rabbits and guinea pigs are composed of **100% calcium carbonate**. Other calculi that are less commonly seen include calcium carbonate mixed with other compounds such as struvite, apatite, and calcium oxalate. When chinchilla get uroliths, they are also typically made of calcium carbonate. Uroliths can be present anywhere in the urinary tract such as nephroliths, ureteroliths, cystoliths, and urethroliths. **Rabbits and guinea pigs are not commonly obstructed, but it can definitely happen in the urethra (particularly in males) and trigone of the bladder for a big cystolith**.

On blood work, general changes can be seen such as anemia (with chronic hematuria), and inflammatory leucograms. Electrolyte and acid-based abnormalities are seen with obstruction.

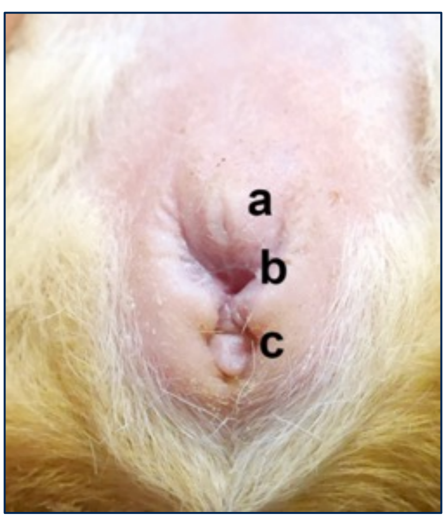
Since calcium carbonate is radio-opaque, survey radiographs are great to diagnose urolithiasis in rabbits and rodents. There is often a single cystolith, especially in guinea pigs. This means that recheck radiographs are not necessary post-cystotomy.

You should **always follow up with an abdominal ultrasound examination** to confirm the location of the urolith, look at the rest of the urinary tract (especially for the presence of hydronephrosis, hydroureter, evidence of cystitis, all of which would not be seen on radiographs), and collect urine for urinalysis and urinary culture. In rare cases, the lith is actually not in the urinary tract. In male guinea pigs, the urolith can migrate into the large seminal vesicles. On culture, when positive (rarely in rabbits, occasionally in guinea pigs), a variety of gram negative bacteria may be isolated and, in guinea pigs, ***Corynebacterium* spp.** (*renale* and *kutscheri*) are frequent isolates (note that it does not induce encrusting cystitis like in dogs).

When animals are obstructed or prior to a cystotomy procedure, they need to have a urinary catheter placed. **The goals are to unobstruct, retropulse stones in the bladder (can be impossible when distal in guinea pigs), and prevent migration of the stones into the urethra prior to surgery (tends to happen with sedation with midazolam and muscle relaxation).** In rabbits, a 3.5-5 Fr soft catheter can easily be placed in males. Females are often not catheterized, but can be catheterized under endoscopic guidance as the urethral orifice is far on the vaginal floor. **Guinea pigs are easy to catheterize. They have a huge urethra so the same size catheter as in rabbits can be used. In males, make sure you stay away from the intromittent sac** (caudal to urethral orifice).

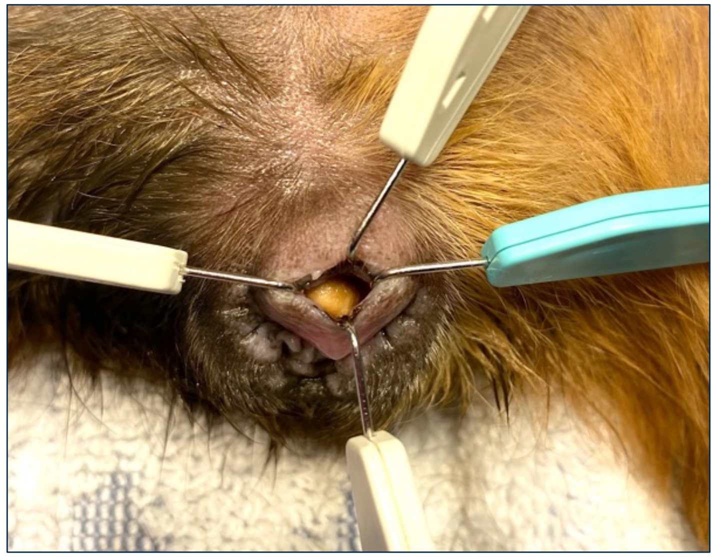


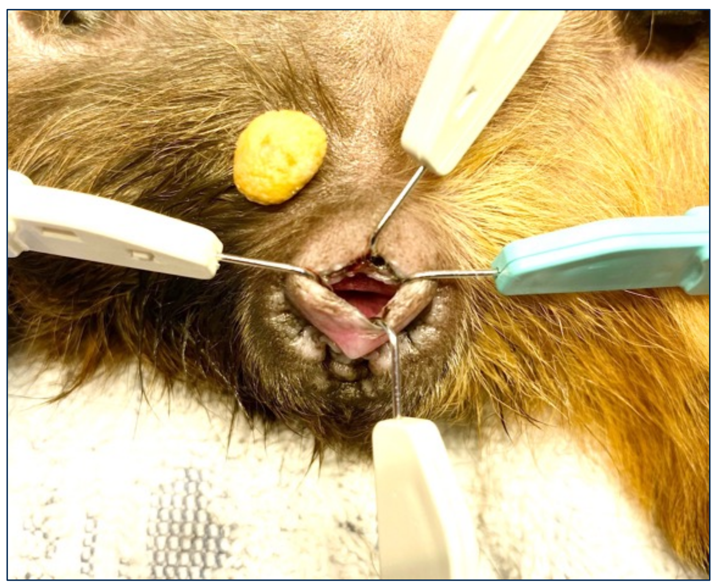
In females, the urethral opening is extravaginal and easy to see (like in all rodents, letter “a” on the following figure). As a result, female guinea pigs are easy to catheterize.



Once placed, the catheter is secured in place using tape and sutures in a butterfly pattern or a Chinese finger trap suture pattern.

Again, **guinea pigs have a huge urethra and sometimes the calculus is almost at the urethral orifice in both males and females**. In males, it can be grasped with forceps if visible or the urethra can be slightly everted to visualize and extract the stone. If slightly more proximal, you may inject some lub in there and then try to milk it out. **In females, a self-retaining retractor (Lonestar) can be placed on the urethral orifice** to allow visualization and extraction of the stone.





Medical therapy is not rewarding so if the stones are big (>5mm), not passed within a certain time frame, and not manually retrievable at the distal urethra, a cystotomy is needed.

The surgery is similar as in other species. If the animal is not catheterize, you may want to catheterize retrogradely to ensure patency of the urethra. **Make sure you leak-test the bladder prior to closing the abdomen**.

The stone can be submitted for analysis, but it is always made of calcium carbonate, so it is not absolutely essential to managing these cases. Again, post-operative radiographs are not needed if there was only a single stone that was removed.

In terms of follow up, you should know that **urolithiasis carry a guarded prognosis in guinea pigs with an overall mortality of up to 50%. Males appear to have poorer prognosis than females, especially if presented with anorexia and weight loss** (likely some hepatic lipidosis also present, as guinea pigs are prone to that with anorexia, unlike rabbits, but this is another story). Guinea pigs also seem to have a higher mortality rate than other species following routine cystotomy, so keep that in mind. Finally, there is a **high recurrence rate in all species** (maybe 10-20%).

For prevention of urolithiasis and recurrence in susceptible animals, the focus is on the diet and water intake. **Owners should attempt to reduce dietary calcium intake by limited high calcium sources such as alfalfa hay and pellets, high calcium greens (kale, parsley, dandelions and others), and hard water. Increasing water intake may also help dilute calcium carbonate crystals.** It has been shown that rabbits prefer to drink from a bowl, but that guinea pigs prefer to drink from a water bottle (probably best to provide both, just in case). No calcium or vitamin supplement should be given (especially to rabbits, which are very susceptible to vit D toxicosis). **Newer studies in guinea pigs have also shown that dietary calcium may not be as big of a factor** as we previously though and that water intake seems to be the most important factor. **Weight loss if overweight and increased exercise may also help.** Urinary acidifiers are poorly effective in these species as their urine is naturally alkaline. **Potassium citrate** is frequently cited as a possible treatment in guinea pigs and in rabbits. It is supposed to work by binding calcium and reducing calciuresis while keeping urine alkaline. It does not seem to be particularly effective, but you are welcome to try.

Nephroliths and ureteroliths are much harder to deal with. Increasing diuresis with intravenous fluids can be tried. Nephrotomy and ureterotomy can also be performed, but are risky and seldom carried out.

Ferrets

You'll be happy to go back to more familiar territory with the mischevious ferret. **The presentation and management are essentially the same as in cats**, so I won't go into a lot of details.

The most common urolith in these naughty mammals used to be struvites. But, twist!!! it is now cystines.

Struvites are associated with the consumption of low quality food for carnivorous animals such as dog food (as you know, the dog is omnivorous, so dog food is not great for ferrets) and food with plant-based proteins that tend to alkalinize urine (and hence promote struvite precipitation).

Cystines have become the most common stones in ferrets over the past 10 years or so. It is **thought to be associated with inbreeding** of the North American ferret population. Some genetic mutations may be altering renal cystine transporters or create a defect in renal tubule reabsorption of cystine, thereby increasing cystinuria. However, genetic investigations have not found any particular mutation yet. **Grain-free diets have also been associated with an increased prevalence.** In Europe, cystine calculi in ferrets are less common. **The big issue is that these stones are radiolucent or mildly radiodense, so they are harder to detect.**

Calcium oxalate is also seen on occasion in ferrets.

The presentation of urolithiasis in ferrets differ from that of rabbits and rodents in that **they pretty much always present obstructed (typically in males).** They also typically have several uroliths.

I'll say it again, the management is identical to cats. You should do a urinalysis and urinary culture, attempt to catheterize the ferret, and perform a cystotomy. Catheterization is straightfoward in the male, but remember that the **urethral opening is not at the tip of the penis, it is ventral and proximal to the "J hook".**

Females are hard to catheterize and a vaginal speculum needs to be used. Perineal urethrostomy can be done in ferrets if the stone cannot be removed from the urethra, but is more challenging to do than in cats because of their small size.

Since ferrets can have different types of stones, it is best to analyze the stones. Dietary changes will depend on the type of stone and urinary pH. In any case, ferrets need to be placed on a high quality protein diet for carnivores.

CYSTITIS

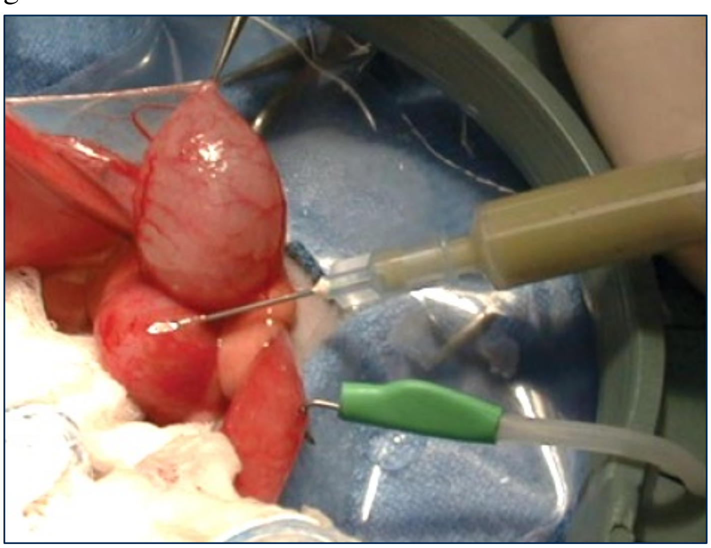
Spontaneous bacterial cystitis is somewhat uncommon in most small mammals commonly seen in practice. However, it does occur with some frequency in guinea pigs, particularly with *Corynebacterium* spp. Always do a urinary culture in all cases of urolithiasis to rule that out.

Cystitis need to be treated based on culture and sensitivity with antibiotics with urinary excretion (such as quinolones).

PROSTATOMEGALY

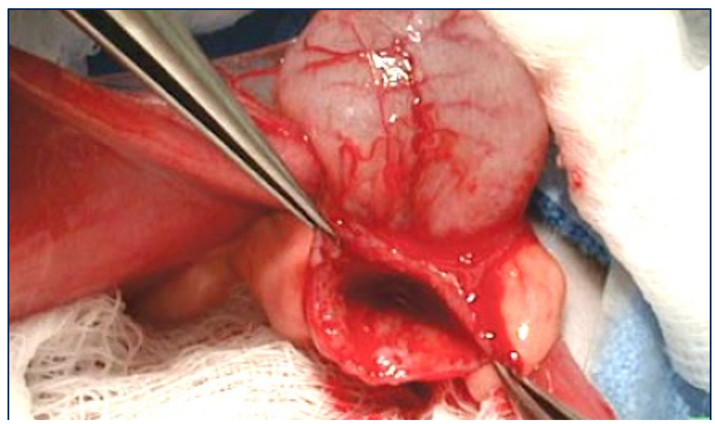
The prostate is technically part of the reproductive system, but in ferrets, it clinically impacts the urinary system primarily, so I put it in this section. It is a problem in male ferrets and it is a **common manifestation of adrenal gland disease** (see endocrine lecture).

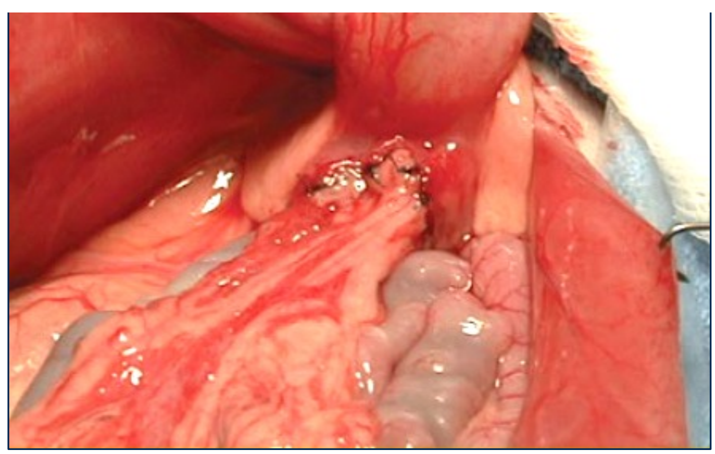
The prostate surrounds the neck of the bladder and proximal urethra in ferrets. As a result, **prostatomegaly frequently leads to urethral obstruction in male ferrets** under androgenic stimulation from the adrenal glands.



Different lesion types may lead to prostatomegaly and urethral occlusion such as **prostatic cysts, prostatic hypertrophy, and prostatitis**. The prostate can easily be visualized by abdominal ultrasound. Treatment initially relies on unblocking the ferret by urethral catheterization. **The prostate can shrink significantly after the injection of leuprolide acetate at higher doses (like 300 mg/kg IM) in a relative short time, such as 24-48h.** So, periodic cystocentesis until the ferret can urinate again is also a good option. **A deslorelin implant can be placed for long-term treatment of adrenal gland disease.** However, it does not work as well as leuprolide for rapidely shrinking prostatic tissues as the onset of action is much slower.

If medical treatment does not work or prostatic cysts are just too big, surgery is indicated. There are **different types of surgical techniques** that have been described such as surgical drainage (could also be done ultrasound-guided but radiologists are reluctant to do it because of the risk of abdominal contamination and peritonitis), marsupialization (making a hole after cleaning the cyst so further fluid will just go into the abdomen, it is controversial as it may lead to peritonitis or focal infection), and **omentalization** (pictures below. Most of the cyst is basically resected and the omentum is packed into the remaining site to provide local healing). Total cyst resection is typically not possible as the urethra is too close.





In any case, antibiotic therapy should also be given in the case of cysts or prostatic abscesses. Lipophilic antibiotics such as TMS are preferred as they penetrate better into prostatic tissue. **Chronic treatment of adrenal gland disease** should also be considered (medical vs surgical).

A very large prostate may also lead to constipation by pressing on the rectum.

URETHRAL PLUGS

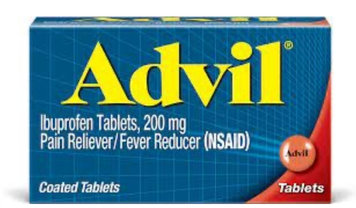
An interesting disease of myomorph rodents (especially rats, mice, gerbils, and hamsters) is the occurence of urethral plugs that may lead to urethral obstruction. It can **sometimes be a normal finding in male rodents, but may also lead to urethral obstruction**. It has the same composition as the vaginal plugs found in females after copulation. It is made of **coagulated secretions from accessory sex glands**, especially the seminal vesicles and the coagulating glands. In some cases, it can be found in the bladder, where it got pushed back through retro-ejaculation. It should not be confused with a urolith. In mice, it is apparently common in the prepuce and is known as the **mouse obstructive genitourinary syndrome**. Surgical opening of the prepucial skin and debridement can be done. Overall, the prevention is by orchidectomy.

RENAL TOXINS

A lot of the same toxins (especially drugs) that are nephrotoxic in other mammals are toxic in small mammals (except grapes that are only toxic in ferrets).

Aminoglycosides are nephrotoxic in most animals and **gentamycin** seems to be the worst. It causes acute tubular necrosis.

Ferrets are very susceptible to ibuprofen toxicity. The minimum lethal dose in them is 220 mg/kg. They are small and inquisitive and love to swallow stuff, so a single tablet of 200 mg could be lethal to a ferret.



It tends to cause pretty **bad gastric ulcers, some neurological signs, and also renal failure**. Ferrets, like cats, are bad at glucuronidation of drugs and many NSAIDs are eliminated by this metabolic pathway, which may explain their increased susceptibility to NSAIDs adverse effects. You will notice that the NSAIDs dose in ferrets is way lower than other small mammals (rabbits and rodents have crazy high doses of meloxicam).

Tiletamine can be nephrotoxic in rabbits. It is part of the tiletamine/zolazepam anesthetic drug combination known as Telazol™. It can cause **nephrosis**, which is irreversible at high doses. This has mainly been demonstrated in New Zealand White Rabbits. However, we don't really use it in pet rabbits because of that. Lead toxicosis may lead (no pun intended) to tubular degeneration in rabbits on top of other signs such as neurological signs.

CHRONIC RENAL DISEASE

Please refer to the corresponding course on renal disease, most of it is applicable to small mammals, so I won't go into a lot of details on the pathophysiology of it.

Introduction

Chronic renal disease is common in all species of small mammals, but **particularly in older ferrets**. The clinical signs are somewhat similar to domestic carnivores, especially in regards to the ferret. So, you will mostly see **PUPD, weight loss, lethargy, anorexia**. In rabbits and rodents, you may see **urine scalding** associated with inappropriate urination. In ferrets, you may see vomiting (remember rabbits and rodents don't vomit) and uremic ulcers as well as pelvic limb lameness (kind of a non-specific sign of disease in them).

The clinical approach is also similar and diagnosis is based on a combination of physical examination (abdominal palpation, palpation of kidneys), blood work (CBC and biochemistry), abdominal ultrasound (shape and size of kidney, echostructure), urinalysis (low USG, especially in dehydrated animals, please **remember that rabbit's urine is frequently iso or hyposthenuric, but should still be concentrated if the patient is dehydrated**), and urinary protein-to-creatinine ratio (to detect and quantify proteinuria). Something that may be more pronounced in rabbits is the severity and onset of renal secondary hyperparathyroidism. In rabbits, as you know, the calcium metabolism is different than other mammals. Renal disease leads to a decrease in renal excretion of calcium, however it is the main mechanism for rabbits to get rid of the high amount of calcium that is passively absorbed from the GI. Therefore, they may be **more prone to ectopic calcification than other animals**. Hypertension and anemia are also other signs that may be seen in animals with chronic renal disease.

Treatment is similar to domestic carnivores such as increasing water intake, chronic administration of supplemental parenteral fluids, phosphate binders in case of hyperphosphatemia (especially in ferrets, less useful in herbivores), and managing complications such as anemia, proteinuria, and hypertension.

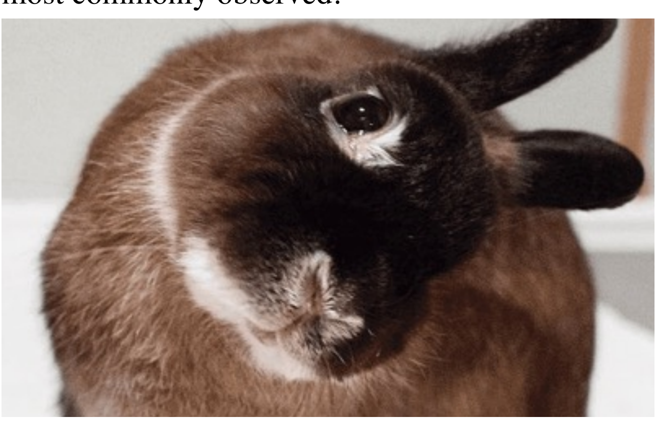
Causes

There are multiple causes that can lead to chronic renal failure, some are common accross mammalian species, and some are more species-specific.

In all species, **interstitial nephritis, fibrosis, and tubular degeneration** are common underlying lesions of renal disease. **Pyelonephritis**, typically associated with ascending infection from the lower urinary tract, are also seen commonly and can cause acute kidney injury. Bacteria that have been implicated include ***Pasteurella multocida*** **in rabbits** (a common culprit for all kinds of infection, especially respiratory and sepsis) and *Staphylococcus* spp. Nephrotoxins (see earlier) may also lead to renal disease. Lastly, neoplasias are fairly uncommon, but will cause renal disease. The most common are lymphoma (particularly common in the ferret) and renal adenocarcinoma.

Some specific considerations when dealing with small mammal species are the following:

-***Encephalitozoon cuniculi:*** see lecture on neurological diseases. *E. cuniculi* is a microsporidian obligate intracellular pathogen (part of the fungi kingdom), which targets the kidneys and neurological system. **Spores are excreted in urine and transmission is urine-oral**. Encephalitozoonosis may lead to nonsuppurative granulomatous nephritis and interstitial fibrosis. Renal impairment is typically chronic or subclinical. Neurological signs such as head tilt are most commonly observed.



The diagnosis is tough as PCR is insensitive on CSF and urine so we **mainly rely on serologic testing**, which does not fully differentiate healthy from infected animals. To complicate things, **seroprevalence is high in the normal rabbit population**. It is also a **zoonosis**, mainly in children. The treatment is fenbendazole or albendazole, but effectiveness is variable. You should also treat the renal disease if present.

-Increased presence of soft tissue calcification in rabbits with renal disease (see explanation earlier).

-**Nephroliths in rabbits and guinea pigs** are very common and hard to deal with. Renal function needs to be estimated in those cases (e.g. excretory urogram).

-**Chronic progressive nephrosis in aging laboratory rats** (especially males of the Sprague-Dawley breed). The disease is characterized by glomerulosclerosis, tubulointerstitial disease, and proteinuria. Dietary factors play an important role and calory restriction and low-protein diets have shown to reduce incidence and severity of the disease.

-**Renal amyloidosis in hamsters**. Chronic deposition of amyloid, which is a by-product of inflammatory proteins.

-Several viruses can cause renal disease in ferrets. **Aleutian disease** (a parvovirus) leads to glomerulonephritis by deposition of immune complexes. The **ferret systemic coronavirus** (similar to FIP in cats) can cause renal lesions.

ACUTE KIDNEY INJURY

Acute kidney injury can be seen in any species of small mammals. With that said, we see and manage it more commonly in rabbits (probably because they are prone to *Pasteurella multocida* sepsis and anesthesia-induced hypotension) and ferrets (probably because they have lots of crazy diseases).

Again, I will refer you to corresponding lectures in small animals as they will be more thorough on the pathophysiology of acute kidney injury, which is important to know for treatment. The only thing that we don't really do is dialysis because of inherent limitations associated with the small size of rabbits and ferrets.

Animals presenting with acute kidney injury are typically **anuric or oliguric**. Causes are similar to other species and include **hypotensive episodes under anesthesia (rabbits and ferrets especially), acute pyelonephritis, nephrotoxins, acute presentation on chronic disease, and renal neoplasia**.

It is very important in those cases to run serial blood gas and electrolyte panels to diagnose and treat abnormalities through fluid therapy and fluid additives. Changes that you should expect include **hyperkalemia (a life-threatening issue), hyperphosphatemia, azotemia, and metabolic acidosis** (typically high anion gap normochloremic metabolic acidosis). **In septic patients, hypoglycemia and ionized hypocalcemia are common.**

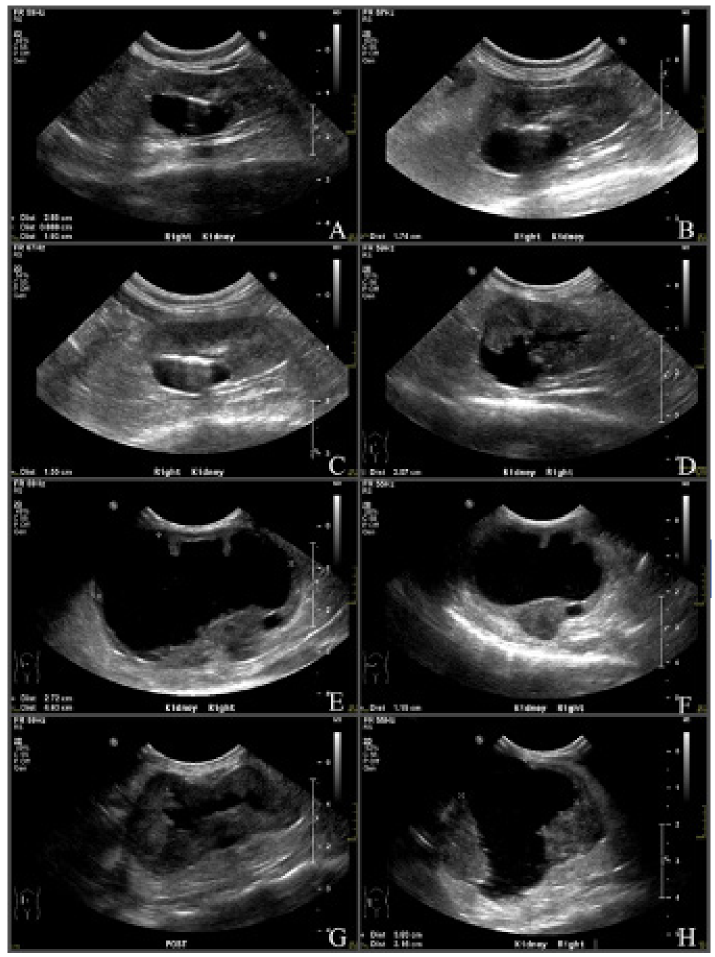
The treatment, again similar as in other species, initially focuses on treating the hypovolemia (or hypovolemic shock). See also the lecture on emergency and critical care. An **initial fluid bolus of 10-20 ml/kg** should be given depending on the degree of hypovolemia (blood pressure, CRT, hydration status and so on). Then **fluid deficits should be corrected.** Lactated Ringer's solution or Plasmalyte-A are the fluids of choice in most situations (NaCl is too acidotic). In most cases, vasopressors will be needed. **Norepinephrine is the drug of choice given as a CRI.** Special note in rabbits: dopamine does not work in them and the dose of norepinephrine is very high at 1 ug/kg/min. Rabbits and ferrets are also prone to fluid overload and many of these shocks, especially septic shock are poorly fluid responsive, so vasopressors are often needed. Cases that responds well to hypovolemic shock therapy and vasopressors will see most of their electrolytic and acid-base abnormalities self-resolve. However, if the acidosis is not resolving, consider giving intravenous **bicarbonates** and consider giving a **CRI of insulin/glucose** for persistent hyperkalemia. Diuretic therapy (furosemide) can also be attempted, but I think it is falling out of favor because of a lack of effectiveness. If sepsis or pyelonephritis are suspected, you can give intravenous antibiotics and IV dextrose supplementation with hypoglycemia. You may elect to also place a urinary catheter to monitor urine production. We only do it in our species after initial treatment and if we think it will be well tolerated as many of these species are hard to catheterize.

RENAL CYSTS IN FERRETS

Renal cysts are **very common in ferrets** and occur with a 10 to 70% prevalence, depending on which studies you look at. These cysts are **mostly incidental** findings, but it is important to know they are common when doing an ultrasonographic examination.

In rare instances, these cysts may lead to clinical issues such as when they are getting too big and disrupt the normal kidney architecture, when neoplastic transformation occurs (see ultrasound below), or with polycystic kidney disease (rare).

The diagnosis is made by ultrasound and treatment is not required in most cases. Fine-needle aspiration of the cysts can be done by ultrasound, but these cysts most often quickly reaccumulate fluids, so we don't really do it.



Rabbits

Urinary incontinence is somewhat common in rabbits. True incontinence should be differentiated from abnormal and improper urination such as occuring with PUPD and pollakiuria, which may be caused by a variety of disorders such as **renal disease, hypothyroidism, psychogenic PUPD (not uncommon in rabbits) and urolithiasis**. In addition, **older rabbits may become reluctant to use their litter box because of spondylosis or arthritis.** True incontinence is linked to neurological disorders (see lecture on neurology) such as **vertebral fractures, IVDD, and *E. cuniculi* infection**.

Hormone-responsive urinary incontinence may occur on occasion in ovariohysterectomized rabbits (same as in dogs, but less common). Treatments include phenylpropanolamine, diethylstilbestrol, and GnRH analogues.

Whatever the cause, urinary scalding may occur in rabbits and sometimes guinea pigs and treatment may be difficult. The treatment of urine scald in rabbits consists of first removing the matted fur, shaving, and cleaning the area.

Be extremely careful when bathing rabbits as they may break their back when they freak out. Then rabbits can be placed on dry clean bedding. **A topical ointment** such as standard wound ointments (SSD for instance) or water-repellent ointments (such as Sudocream™) may be used. **Antibiotics** should be given for the pyoderma as well as **analgesics**. **Most severe cases may need a urinary catheter** to be maintained in order to prevent further soiling of the perineum with urine while the skin is healing.

Some of these cases can be pretty bad with severe pyoderma. Rabbits housed outside may also suffer from myiasis of the area, which further complicate things. Last, you need to treat the underlying disease or give chronic pain meds (arthritis for instance).

Ferrets

Likewise, in ferrets, true incontinence must be differentiated from just chronic renal disease, PUPD, and urolithiasis. Notable causes of neurological incontinence in ferrets include **vertebral fracture, IVDD, aleutian disease, rabies** (rare, but incontinence is very common in experimental rabies infection in ferrets), and **vertebral metastasis** of a variety of tumors (multiple myeloma, lymphoma typically). Ferrets develop a lot of cancers, so it is more of a differential in them than in rabbits.

MISCELLANEOUS

There are many diseases of the urinary tract that we have not covered because they are less common and I just want you to focus on what you are most likely to initially see in practice. However, one notable disease that is kind of unique to rabbits that is worth knowing is **urinary bladder herniation, typically in the scrotum or inguinal area of males**. It is likely due to the large open inguinal rings of males. This should be repaired surgically.

Another thing that is kind of bewildering in ferrets is the occurrence of **paraurethral cysts associated with**

hormonal stimulation due to adrenal gland disease. They come from a hyperplasia of mesonephric or paramesonephric ducts or paraprostatic tissues. When large enough, they may lead to urinary obstruction. Treatment is also surgical, but they can be drained for temporary relief by ultrasound guided aspiration.

The Ten Most Important Facts

1. Rabbits passively absorb dietary calcium from their instestines unregulated by vitamin D and excrete the surplus massively through the urine. Guinea pigs also absorb more calcium from their diet and have high urinary fractional excretion of calcium.
2. Urine scald (aka hutch burn, perineal dermatitis) is commonly seen in rabbits (and to a lesser measure in guinea pigs) with a wide variety of urinary tract disorders or pain in older rabbits. This can lead to severe pyoderma.
3. Urine appearance in rabbits is frequently cloudy because of "sludge". The specific gravity is frequently low as they don't concentrate their urine much and pH is high (like in most herbivorous species).
4. Calcium carbonate is the main urolith in rabbits and guinea pigs. Management is frequently surgical. Post-operatively, dietary calcium should be reduced (no alfalfa hay or pellets, limit high calcium greens and hard water) and water intake should be encouraged.
5. Guinea pigs have a poorer prognosis with urolithiasis than other mammalian species, especially in anorexic male. Recurrence is common. Distal urethral stones can often be manually removed without the need for surgical extraction in both males and females.
6. Ferrets' most common urolith used to be struvites (because of poor quality carnivorous food), but, in an incredible turn of events, it is now cystines, which are radiolucent and frequently cause urinary obstruction. The cause of cystine stones is thought to be genetic and/or nutritional (grain-free diet).
7. Prostatomegaly in ferrets, caused by prostatic hypertrophy, prostatic cysts, or prostatitis, is associated with adrenal gland disease and may lead to urethral occlusion. Often times, a single injection of a high dose of leuprolide acetate will unobstruct male ferrets. Surgical omentalization of problematic cysts or abscesses is also possible.
8. Ibuprofen is very toxic to ferrets. A single pill could lead to a ferret's tragic demise. It also causes gastric ulceration and neurological signs. Ferrets are just not good at the good ol' glucuronidation of NSAIDs.
9. *Encephalitozoon cuniculi* is a common microsporidian infection of rabbits that targets the kidney and neurological system. It may lead to chronic or subclinical renal impairment.
10. Renal cysts are common in ferrets and typically incidental. However, some cases of neoplastic transformation have occurred and very large cyst or multiple cysts may disrupt renal architecture.